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LOT-01 — COMPONENTS BUYING LIST (BOM) + 100-UNIT COSTING
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DOCUMENT    HW-02 / BOM
ISSUE       2026.11.12 · REV A
CURRENCY    USD, planning estimates at qty 100–110
NOTE        Supplier sites were unreachable this session. Prices are
            estimates from known market ranges. Verify price + stock
            before every order. Machine-readable copy:
            hardware/bom/LOT-01-BOM-revA.csv
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A · PROTOTYPE KIT (phase P1 — buy this first, ×3 sets)

Off-the-shelf boards to prove the full loop (notification → screen, COPY → Log tab, weather → server, camera frame → server) before any custom PCB.

#	Part	Role	Est. $	Link (verify)
1	Seeed Studio XIAO ESP32S3 Sense	MCU + camera + 8 MB PSRAM	14	https://www.seeedstudio.com/XIAO-ESP32S3-Sense-p-5639.html
2	Adafruit SHARP Memory Display 1.3" 168×144	Screen (same family as product)	25	https://www.adafruit.com/product/3502
3	Adafruit BME688 breakout	Weather + air (AI gas)	20	https://www.adafruit.com/product/5046
4	SparkFun 6DoF IMU BMI270 (Qwiic)	Motion / wake gesture	18	https://www.sparkfun.com/products/22397
5	Adafruit Universal Qi Receiver Module	Wireless charge test	15	https://www.adafruit.com/product/1901
6	Adafruit Qi Wireless Charging Transmitter 5V	Dock stand-in	30	https://www.adafruit.com/product/2162
7	LiPo 3.7 V 150 mAh	Battery	6	https://www.adafruit.com/product/1317
8	Vibrating mini motor disc	Pager buzz	2	https://www.adafruit.com/product/1201
9	6 mm tactile switch + jumper wires + perfboard	COPY button	5	any
	Per set ≈ US$ 135 · ×3 sets ≈ US$ 405			

B · PRODUCTION BOM — DEVICE (per unit, qty 110)

Ref	Part (manufacturer P/N)	Package / size	Qty	Est. $	Link (verify)
U1	Espressif **ESP32-S3-PICO-1-N8R2** (SiP: MCU + 8 MB flash + 2 MB PSRAM + crystal)	LGA 7.0 × 7.0 × 1.0 mm	1	3.60	https://www.espressif.com/en/products/socs/esp32-s3
ANT	Johanson **2450AT18A100** 2.4 GHz chip antenna	0603-size, 1.6 × 0.8 mm	1	0.45	https://www.digikey.com/en/products/result?keywords=2450AT18A100
DS1	Sharp **LS013B7DH03** 1.28" memory-in-pixel LCD, 128 × 128, reflective	~ 0.75 mm thick (verify)	1	10.00	https://www.digikey.com/en/products/result?keywords=LS013B7DH03
CAM	GalaxyCore **GC0308** VGA camera, FPC module, ≤ 3.0 mm Z	6 × 6 mm class	1	3.00	https://www.lcsc.com/search?q=GC0308
U2	Bosch **BME688** gas / T / RH / P sensor	LGA 3.0 × 3.0 ×	1	8.50	https://www.bosch-sensortec.com/products/environmental-sensors/gas-

Ref	Part (manufacturer P/N)	Package / size	Qty	Est. $	Link (verify)
	(AI-Studio)	0.93 mm			sensors/bme688/
U3	Bosch **BMI270** 6-axis IMU	LGA 2.5 × 3.0 × 0.83 mm	1	2.40	https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi270/
U4	TI **BQ51050B** Qi receiver + Li-ion charger	DSBGA / VQFN	1	2.60	https://www.ti.com/product/BQ51050B
L1	Würth **WE-WPCC** Qi receiver coil, ≤ 25 mm, ≤ 0.6 mm, with ferrite (pick from series)	flat	1	3.20	https://www.we-online.com/en/components/products/WE-WPCC
U5	TI **TPS62840** buck 3.3 V, 750 mA, 60 nA Iq	WSON / DSBGA	1	1.20	https://www.ti.com/product/TPS62840
U6	Analog Devices **MAX17048** fuel gauge	WLP 0.9 × 1.7 mm	1	1.80	https://www.analog.com/en/products/max17048.html
BT1	LiPo cell 2.0 × 25 × 30 mm, ~ 90 mAh, PCM, UN38.3 (DVT: 1.6 mm cell)	pouch	1	2.80	https://www.digikey.com/en/products/result?keywords=lipo%20303030
SW1	C&K **KMR2** series tactile switch (COPY)	4.2 × 2.8 × 1.9 mm	1	0.40	https://www.ckswitches.com/products/switches/product-details/Tactile/KMR2/
M1	Coin ERM vibration motor 8 mm × ≤ 2.0 mm (verify thickness)	coin	1	0.70	https://www.digikey.com/en/products/result?keywords=coin%20vibration%20motor%208mm
Q1	N-MOSFET for motor + flyback diode	SOT-523 / SOD-923	1	0.15	—
RT1	NTC 10 k 0402 (battery temp → BQ51050B TS)	0402	1	0.05	—
—	Passives (≈ 45 × 0201/0402 R/C/L)		45	0.90	—
VNT	GORE adhesive protective vent (for BME688)	Ø 3–4 mm	1	0.60	https://www.gore.com/products/categories/venting
PCB	6-layer HDI, 0.6 mm (DVT: 0.4 mm), 36 × 36 mm, ENIG, via-in-pad	PCBWay	1	7.50	https://www.pcbway.com/orderonline.aspx
PCBA	SMT assembly (both sides), stencil, X-ray for LGA	PCBWay	1	11.00	https://www.pcbway.com/pcb-assembly.html
GLS	Cover glass 0.55 mm aluminosilicate, 38 × 38 mm, black ink mask, clear camera + screen windows, AR coat	custom	1	4.50	glass vendor (RFQ)
BDY	316L body, 2 parts, CNC + mirror polish (back), bead-blast (bezel, optional)	PCBWay CNC	1 set	34.00	https://www.pcbway.com/rapid-prototyping/
BTN	316L COPY button cap, CNC, polished	Ø 5 mm	1	1.50	PCBWay CNC
GSK	0.3 mm PEEK seam gasket (RF window + seal)	laser cut	1	0.60	PCBWay / local
SCR	M1.2 × 3 Torx, A4 stainless		4	0.25	—

Ref	Part (manufacturer P/N)	Package / size	Qty	Est. $	Link (verify)
ADH	tesa 61395 / 3M 5925 die-cut adhesive (glass, battery)		1 set	0.40	—
	Device subtotal ≈ US$ 102 / unit				

C · PRODUCTION BOM — LOT DOCK (wireless charger, item 12 + 19)

Part	Qty	Est. $	Note
Qi 5 V transmitter PCBA (single-coil, A11-type, FOD)	1	4.00	Off-the-shelf for run 1; custom TX in run 2
316L CNC puck ring Ø 50 × 8 mm, polished	1	12.00	PCBWay CNC; must **not** cover the coil
Top plate 1.0 mm ceramic or glass Ø 46 mm	1	2.50	Coil sits directly under it
Silicone foot ring + 2 × M2 screws	1	0.60	
USB-C cable 1 m, braided, white or silver	1	2.50	
Dock subtotal ≈ US$ 22 / unit			

D · 100-UNIT RUN — COST ROLL-UP (order 110: 100 + 10 spares)

Line	Per unit	× 110
Device BOM + PCBA + body	102	11,220
LOT Dock	22	2,420
Packaging (box, foam, printed manual card)	5	550
Shipping, duties, freight (est.)	6	660
Recurring subtotal	**135**	**14,850**
NRE: stencils, CNC fixtures, polish jigs	—	1,200
NRE: EVT (10) + DVT (20) builds	—	6,500
NRE: test jig (pogo bed + firmware)	—	900
Certification FCC + CE (radio, pre-scan + test)	—	8,000–15,000
Program total		**≈ 31,500 – 38,500**
All-in cost per delivered unit (÷ 100)		**≈ 315 – 385**

Recurring cost drops to about **US$ 110–125** per unit at 1,000 units; the biggest one-time line is certification. Using a pre-certified ESP32-S3 **module** (not the SiP) can cut certification cost, but it is 2.4–3.2 mm thick. That trade is open until the EVT Z-stack is measured.

E · ORDER SEQUENCE

1. **Now:** Prototype kit A × 3 (≈ US$ 405).
2. **After P1 passes:** EVT — 10 × PCB/PCBA + 10 × CNC body at 6.0 mm (PCBWay), 10 × cover glass, 12 × battery (≈ US$ 2,500).
3. **After EVT passes:** DVT — 20 units at 5.0 mm + 20 Docks.
4. **After DVT passes + cert pre-scan:** PVT — 110 units.

Long-lead items to quote early: cover glass (custom), LS013B7DH03 (stock varies), thin LiPo with UN38.3 report (needed to ship by air).